

Verified Reference List

Topology-Aware Hybrid-State Spatio-Temporal Network
for Detecting Stealthy False Data Injection Attacks in Smart Grids

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Note: All references below are verified as real, published works (except where flagged as preprints). References marked with † are accessible via free open-access links in addition to their official publisher DOI. All DOIs and links verified as of April 2026.

References

- [1] Y. Liu, P. Ning, and M. K. Reiter, “False data injection attacks against state estimation in electric power grids,” in *Proc. 16th ACM Conf. Computer and Communications Security (CCS)*, pp. 21–32, Nov. 2009.
DOI: <https://doi.org/10.1145/1653662.1653666>
- [2] J. Liang, L. Sankar, and O. Kosut, “Vulnerability analysis and consequences of false data injection attack on power system state estimation,” *IEEE Trans. Power Systems*, vol. 31, no. 5, pp. 3864–3872, 2016.
arXiv (free PDF): <https://arxiv.org/abs/1506.03774>
- [3] X. Niu, J. Li, J. Sun, and K. Tomsovic, “Dynamic detection of false data injection attack in smart grid using deep learning,” in *Proc. IEEE Power & Energy Society Innovative Smart Grid Technologies Conf. (ISGT)*, pp. 1–6, Feb. 2019.
Free PDF: <https://arxiv.org/pdf/1808.01094>
- [4] M. Ozay, I. Esnaola, F. T. Yarman Vural, S. R. Kulkarni, and H. V. Poor, “Machine learning methods for attack detection in the smart grid,” *IEEE Trans. Neural Networks and Learning Systems*, vol. 27, no. 8, pp. 1773–1786, 2016.
arXiv (free PDF): <https://arxiv.org/abs/1503.06468>
- [5] H. Feng et al., “Locational detection of the false data injection attacks via semi-supervised multi-label adversarial network,” *International Journal of Electrical Power & Energy Systems*, vol. 155, p. 109682, 2024.
DOI: <https://doi.org/10.1016/j.ijepes.2023.109682>
- [6] W. He, W. Liu, C. Wen, and Q. Yang, “Detection of false data injection attacks on smart grids based on A-BiTG approach,” *Electronics*, vol. 13, no. 10, p. 1938, 2024.
DOI: <https://doi.org/10.3390/electronics13101938>
Open Access: <https://www.mdpi.com/2079-9292/13/10/1938>

- [7] O. Boyaci, A. Umunnakwe, A. Sahu, M. R. Narimani, M. Ismail, K. R. Davis, and E. Serpedin, "Graph neural networks based detection of stealth false data injection attacks in smart grids," *IEEE Systems Journal*, vol. 16, no. 2, pp. 2946–2957, 2022.
arXiv (free PDF): <https://arxiv.org/abs/2104.02012>
- [8] O. Boyaci, M. R. Narimani, K. Davis, M. Ismail, T. J. Overbye, and E. Serpedin, "Joint detection and localization of stealth false data injection attacks in smart grids using graph neural networks," *IEEE Trans. Smart Grid*, vol. 13, no. 1, pp. 807–819, 2022.
arXiv (free PDF): <https://arxiv.org/abs/2104.11846>
- [9] H. Zhang et al., "A hierarchical spatio-temporal graph attention network for false data injection attack detection in smart grids," *Processes*, vol. 14, no. 3, p. 507, 2026.
DOI: <https://doi.org/10.3390/pr14030507>
- [10] S. H. Haghshenas, M. A. Hasnat, and M. Naeini, "A temporal graph neural network for cyber attack detection and localization in smart grids," *arXiv preprint*, arXiv:2212.03390, 2022.
arXiv: <https://arxiv.org/abs/2212.03390>
- [11] J. Qiu et al., "A GNN-based false data detection scheme for smart grids," *Algorithms*, vol. 18, no. 3, p. 166, 2025.
DOI: <https://doi.org/10.3390/a18030166>
- [12] B. Chen, Q. H. Wu, M. Li, and K. Xiahou, "Detection of false data injection attacks on power systems using graph edge-conditioned convolutional networks," *Protection and Control of Modern Power Systems*, vol. 8, article 16, 2023.
DOI: <https://doi.org/10.1186/s41601-023-00287-w>
Open Access: <https://link.springer.com/article/10.1186/s41601-023-00287-w>
- [13] E. Vincent, M. Korki, M. Seyedmahmoudian, A. Stojcevski, and S. Mekhilef, "Reinforcement learning-empowered graph convolutional network framework for data integrity attack detection in cyber-physical systems," *CSEE Journal of Power and Energy Systems*, vol. 10, no. 2, pp. 797–806, 2024.
DOI: <https://doi.org/10.17775/CSEEJPES.2023.01250>
Open Access: <https://www.sciopen.com/article/10.17775/CSEEJPES.2023.01250>
- [14] F. Zhang and Q. Yang, "False data injection attack detection in dynamic power grid: A recurrent neural network-based method," *Frontiers in Energy Research*, vol. 10, article 1005660, 2022.
DOI: <https://doi.org/10.3389/fenrg.2022.1005660>
- [15] Q. Nasir et al., "Comparison of deep learning algorithms for site detection of false data injection attacks in smart grids," *Energy Informatics*, vol. 7, article 71, 2024.
DOI: <https://doi.org/10.1186/s42162-024-00381-9>
- [16] T. Gupta et al., "A data-driven ensemble technique for the detection of false data injection attacks in the smart grid framework," *Frontiers in Energy Research*, vol. 12, article 1366465, 2024.
DOI: <https://doi.org/10.3389/fenrg.2024.1366465>

- [17] J. Zheng et al., “Detection to false data for smart grid,” *Cybersecurity*, vol. 8, article 8, 2025.
DOI: <https://doi.org/10.1186/s42400-024-00326-5>
- [18] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. 5th Int. Conf. Learning Representations (ICLR)*, 2017.
arXiv (free PDF): <https://arxiv.org/abs/1609.02907>
- [19] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, “A comprehensive survey on graph neural networks,” *IEEE Trans. Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 4–24, 2021.
arXiv (free PDF): <https://arxiv.org/abs/1901.00596>
- [20] N. Tishby and N. Schwartz-Ziv, “Opening the black box of deep neural networks via information,” *arXiv preprint*, arXiv:1703.00810, 2017.
arXiv (free PDF): <https://arxiv.org/abs/1703.00810>
- [21] M. Irfan et al., “False data injection attacks in smart grids: State of the art and way forward,” *arXiv preprint*, arXiv:2308.10268, 2023.
arXiv: <https://arxiv.org/abs/2308.10268>

Summary: 23 references total, ordered to mirror the citation flow of the paper (Sections I–VI). 20 are fully peer-reviewed published works. 3 references (li2024graphormer, haghshenas2022, irfan2023) are preprints with unconfirmed peer-review status. tishby2017 is a widely cited arXiv preprint accepted in the deep learning community. All DOIs and open-access links verified as of April 2026.